

Course Code : ECT 258

WYXZ/MS-23/1755

Fourth Semester B. Tech. (Electronics and Communication Engineering) Examination

MICROPROCESSORS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions carry marks as indicated.
- (2) Solve any **Three** questions from each **Part**.
- (3) Due credit will be given to neatness and diagrams.
- (4) Use of electronic gadgets leaving scientific calculator is not allowed.
- (5) Assume suitable data wherever necessary.

PART A

1. (a) Which are the basic control signals of 8085 ? How do you generate $\overline{MEMR}$, $\overline{MEMW}$, $\overline{IOR}$ and $\overline{IOW}$ signals from basic signals of 8085 ?
5(CO1)
- (b) Explain how stack is shared by microprocessor and programmer. 5(CO1)
2. (a) A system needs 4 KB of EPROM and 4 KB of RAM. ROM address starts at 0000H and RAM address starts at 2000H. The available chips are :
 - (i) 2716 EPROM (2 K × 8) with necessary address, data, read and chip select lines.
 - (ii) 6116 RAM (2 K × 8) with necessary address, data, control and chip select lines. 5(CO3)
- (b) Describe clearly the steps that takes place when the INTR pin receives a pulse and illustrate hardware interrupts RST 0, RST 2, RST 7.5 and RST 6.5.
5(CO3)

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3. (a) An 8085-8255 based microprocessor is required to interface 8 switches and seven segment display. Also write a program to display the switch number, (if a switch is pressed) on 7-segment display. 7(CO3)
- (b) What are the various modes of operation of 8255 ? Also give the control word format. 3(CO2,3)
4. (a) State similarities and differences between CALL and RET Vs. PUSH and POP instruction of 8085. 5(CO2)
- (b) Explain the meaning of following instructions of 8085 microprocessor :
 - (i) DAD H
 - (ii) RAR
 - (iii) XTHL
 - (iv) CMP B
 - (v) INR M
 5(CO2)

PART B

5. (a) Explain the architecture of 8086 microprocessor. 7(CO1)
- (b) Recall pins of 8086 microprocessor and explain any **Three** the significance of following pins in brief :
 - (i) $\overline{BHE}$ / S7
 - (ii) Queue status QS1, QS0
 - (iii) $\overline{DEN}$ – Data Enable
 - (iv) Status lines $\overline{S2}$, $\overline{S1}$, $\overline{S0}$
 - (v) RESET
 3(CO1)
6. (a) Explain the working and need of interrupts in 8086 microprocessor using example. 5(CO1,3)

- (b) Give explanation of any **Two** instruction of 8086 and its addressing modes :
- (i) MOV AX, [BX] [SI]
 - (ii) ADD 50H [BX] [SI], BP
 - (iii) XOR AX, AX 5(CO2)
7. (a) Recall and explain the important features of Intel 286 and 386 microprocessor. 5(CO4)
- (b) Explain the significance of real and virtual memory. 5(CO4)
8. (a) Decide whether the parity of a given number is even or odd. If parity is even set DL to 00, else set DL to 01. The given number may be a multibyte number. 5(CO4,5)
- (b) Write an assembly language program using 8085 to convert an ASCII digit stored at A200H into its binary equivalent. Store the binary equivalent at the same location. 5(CO4,5)

